

Title : Project Registration & Progress Review

FF No. 180

Department: Computer Engineering	Academic Year: 2026-2027
Semester : 5	Group No. : TY CS-K-2
Project Title: Pulse AI: An Intelligent Timesheet and Invoicing Assistant (Sponsored By Emerson Electric & Co)	
Project Area: Artificial Intelligence / Machine Learning (Enterprise Automation)	

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Project Synopsis

1. Project Title

Project Pulse AI: An Intelligent Timesheet and Invoicing Assistant.

2. Introduction / Overview

Project Pulse AI is an artificial intelligence powered platform designed to automate the validation of employee timesheets, the analysis of work hours, and the preparation of payroll and invoicing data. The system is intended to reduce the manual workload currently placed on Human Resource (HR) and Finance departments while simultaneously improving the accuracy, consistency, and transparency of the entire timesheet to invoice workflow.

3. Problem Statement

In most organizations, employee timesheets are still reviewed manually before payroll processing and client invoicing can take place. HR and Finance personnel are required to spend a considerable amount of time verifying logged working hours, identifying inconsistent or missing entries, approving submissions, and subsequently preparing invoices based on this data. Because this process is repetitive, largely manual, and dependent on human judgment, it is highly susceptible to error. Such errors frequently result in delayed employee payments, incorrect client billing, and an overall reduction in departmental efficiency. There is, therefore, a clear need for an intelligent, automated solution that can validate timesheet data reliably and assist staff in generating accurate, payroll ready information with minimal manual intervention.

4. Objectives

- 1) To automate the process of timesheet validation and reduce dependency on manual review.
- 2) To detect missing, incomplete, or inconsistent work hour entries in real time using artificial intelligence.
- 3) To assist HR and Finance teams in preparing payroll data and generating client invoices.
- 4) To generate AI driven insights and summaries of employee work patterns for managerial review.
- 5) To reduce the overall manual effort involved in the payroll and invoicing cycle while improving accuracy and transparency.

5. Proposed Methodology

The proposed system will be developed in a phased manner, beginning with requirement analysis and database design, followed by the implementation of the core validation engine and the user facing dashboards. The broad methodology to be followed is outlined below.

- 1) Data Collection: Employees will submit their daily or weekly working hours through a web based submission interface.

- 2) AI Based Validation: The submitted entries will be analyzed using a machine learning or large language model based engine to flag anomalies such as missing hours, duplicate entries, or unusual patterns.
- 3) Approval Workflow: Flagged and unflagged entries will be routed through an automated approval workflow for managerial sign off.
- 4) Payroll and Invoice Preparation: Once approved, validated work hour data will be aggregated and formatted into payroll ready summaries and client invoices.
- 5) Reporting and Analytics: Dashboards will be provided for employees, managers, HR, and Finance to monitor submissions, approvals, and generated insights.
- 6) Testing and Deployment: The system will be tested for accuracy of validation logic, security of data, and usability, prior to deployment on a cloud platform.

7. Key Features

- 1) Employee timesheet submission portal.
- 2) AI based validation and anomaly detection.
- 3) Automated multi level approval workflow.
- 4) Payroll ready work hour summaries.
- 5) Invoice generation assistance.
- 6) Role based dashboards and analytics for employees, managers, HR, and Finance.
- 7) Automated notifications and reminders.

8. Proposed Technology Stack

Frontend: React or Next.js

Backend: Node.js with Express

Database: PostgreSQL or MySQL

Artificial Intelligence: Large Language Model or OpenAI API based engine for validation, summarization, and assistant features

Authentication: JWT or OAuth

Cloud Hosting: AWS, Azure, Render, or Vercel

Version Control: Git and GitHub

9. Expected Outcomes and Benefits

- 1) Faster and more reliable payroll processing.
- 2) Reduced manual verification effort for HR and Finance staff.
- 3) Fewer billing and invoicing errors.
- 4) Increased transparency for employees regarding their logged hours.
- 5) Improved overall productivity of HR and Finance teams.

10. Future Scope

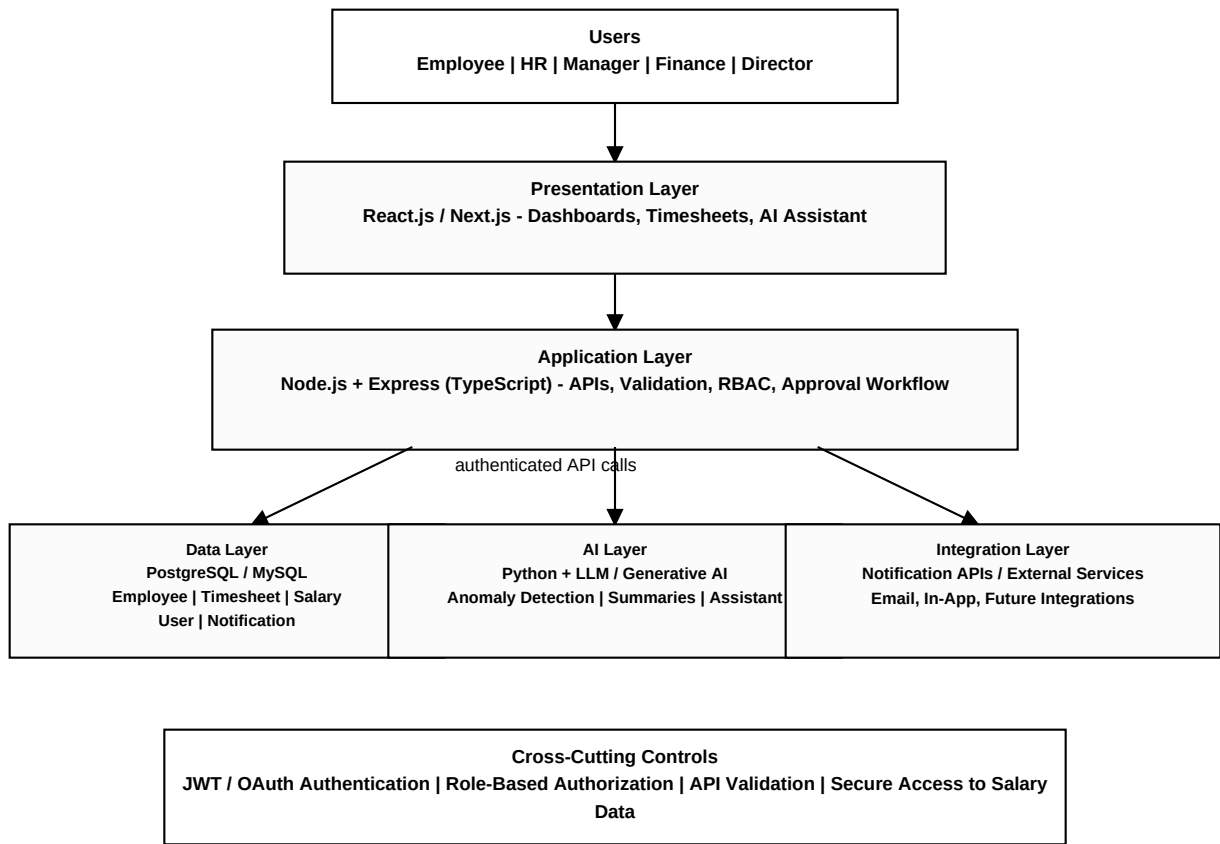
The system may be extended to integrate with established Enterprise Resource Planning (ERP) platforms such as SAP or Oracle, connect with existing biometric or digital attendance systems,

6. System Architecture

The proposed system follows a layered web application architecture in which the user interface, application services, data layer, and AI services communicate through authenticated APIs. The architecture supports employee, HR, manager, Finance, and Director roles with role-based access control.

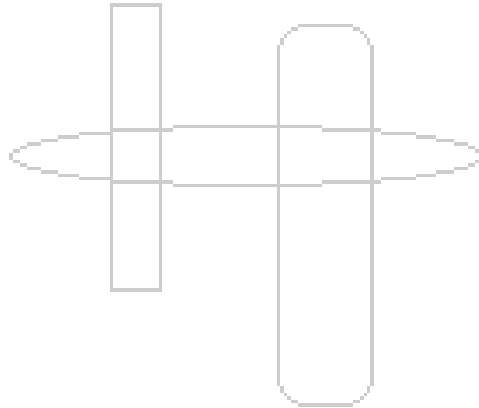
- Presentation Layer: React.js / Next.js provides timesheet submission, role-based dashboards, analytics, and the AI assistant interface.
- Application Layer: Node.js with Express exposes APIs, validates requests, applies JWT/OAuth authorization, and manages timesheet and approval workflows.
- Data Layer: SQL (PostgreSQL / MySQL) stores employee, timesheet, salary, user-role, and notification information.
- AI Layer: Python-based AI services and a Generative AI / LLM engine support anomaly detection, summaries, intelligent queries, and assistant features.
- Integration Layer: APIs connect notification services and future external systems while preserving authorization boundaries.

System Architecture Diagram



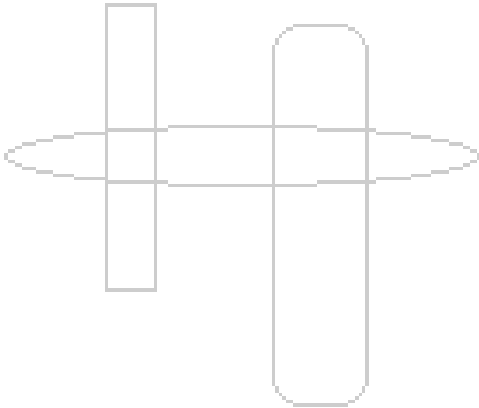
Data and service requests pass through the application layer so that role permissions are enforced before employee or salary information is returned. The architecture can scale by adding employees, departments, managers, and organizational units without changing the core application structure.

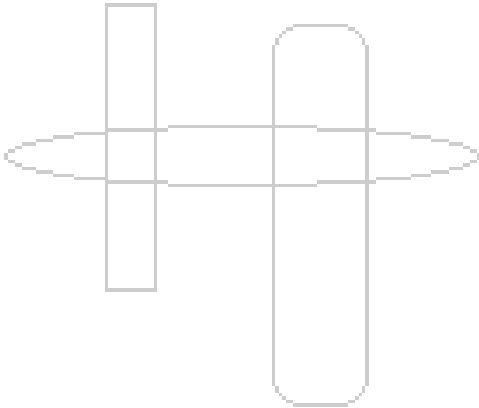
incorporate predictive workforce analytics, support a multilingual AI assistant, and offer a dedicated mobile application for on the go access by employees and managers.



Group No.			
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	Group No.:	Date:
Progress Review Report		
		
Signature of Guide:		

Review No.: 2	Group No.:	Date:
Progress Review Report 		
Signature of Guide:		